Circuit theory – from Kirchhoff to Bode

Circuit theory is the grammar of electrical engineering. With two laws and three kinds of component one can describe everything from a torch to a loudspeaker crossover – and through the analogies, mechanical and acoustic systems too. At DTH a course in elementary circuit theory went together with a laboratory course in circuits and electronics. The page has an interactive RLC circuit with frequency response and step response.

> DTH, course 4901 (1986), 6 points, and course 4932 (1985), 3 points: Elementary circuit theory and Exercises in circuit theory and electronics. The laboratory course gave the practical skills – oscilloscope, function generator, multimeter and breadboard – on which all later laboratory work built.

Ohm and Kirchhoff

Georg Simon Ohm found in 1827 that the current through a conductor is proportional to the voltage: $U = RI$. Gustav Kirchhoff formulated in 1845 – as a 21-year-old student – the two laws which, together with the component equations, determine everything in a circuit:

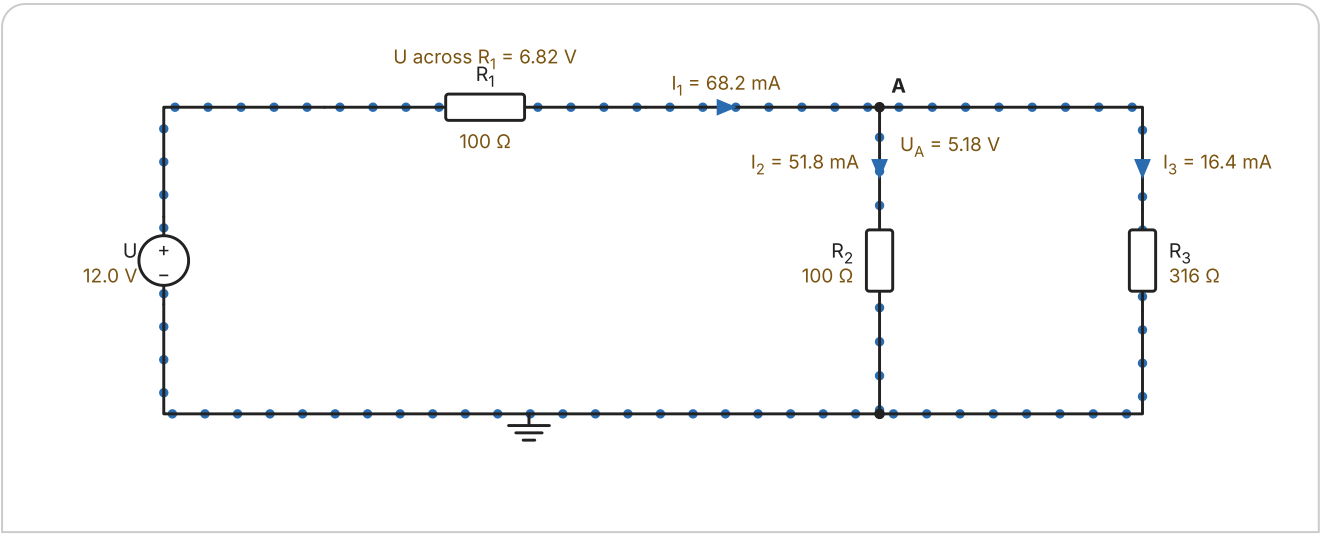
The current law

The sum of currents into a node is zero: $\sum I_k = 0$. That is conservation of charge.

The voltage law

The sum of voltages around a closed loop is zero: $\sum U_k = 0$. That is conservation of energy – the electric field in a static circuit is conservative.

Applied systematically, the laws give nodal analysis: choose a reference node, give each node an unknown potential and write the current law for each node. The result is a linear system $\mathbf{G}\,\mathbf{u} = \mathbf{i}$, which a computer solves in an instant. That is exactly what the circuit simulator SPICE (Berkeley, 1973) does.



Thévenin and Norton

Any linear circuit seen from two terminals can be replaced by a voltage source U_{Th} in series with a resistance R_{Th} (Léon Thévenin, 1883) – or by a current source in parallel with the same resistance (Edward Norton, 1926). U_{Th} is the open-circuit voltage, and R_{Th} the ratio of open-circuit voltage to short-circuit current. The theorem makes it easy to see what happens when a load is connected. Maximum power is delivered to the load when $R_L = R_{\mathrm{Th}}$ – *power matching* – but then half the power is lost in the source. That is why power stations and loudspeaker amplifiers have a low internal resistance, while antennas and transmission lines are matched.

Capacitor and inductor

A capacitor stores energy in an electric field, an inductor in a magnetic one:

$$i = C\frac{du}{dt}, \qquad u = L\frac{di}{dt}, \qquad W_C = \tfrac{1}{2}Cu^2, \qquad W_L = \tfrac{1}{2}Li^2$$

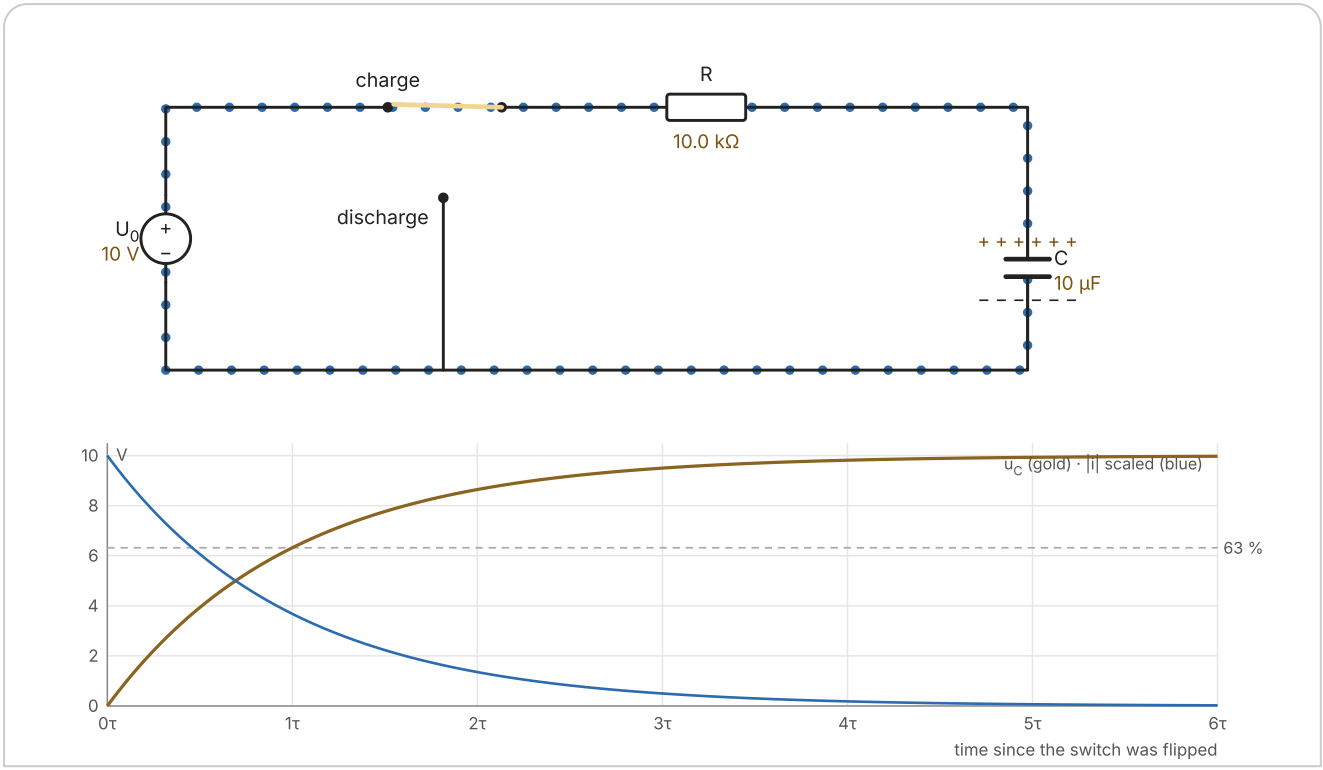
The voltage across a capacitor and the current through an inductor cannot jump – that would require infinite power. This is the key to all switching phenomena: the spark when the current to a coil is broken (a car's ignition coil exploits it), and the current surge when an uncharged capacitor is switched on.

Transients

When a voltage U_0 is connected to a resistor in series with a capacitor, the capacitor charges exponentially with the time constant $\tau = RC$: $u_C(t) = U_0(1 - e^{-t/\tau})$. After 5τ it is 99 % charged. An RLC circuit gives a second-order differential equation,

$$LC\frac{d^2u_C}{dt^2} + RC\frac{du_C}{dt} + u_C = u(t), \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \qquad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

With $\zeta < 1$ the circuit is underdamped and oscillates before settling; with $\zeta > 1$ it is overdamped and creeps slowly up; $\zeta = 1$ is critical damping – the fastest response without overshoot, aimed for in measuring instruments and car shock absorbers. It is the same equation as for a damped pendulum and for a loudspeaker in a box (see electroacoustics).



Phasors and complex impedances

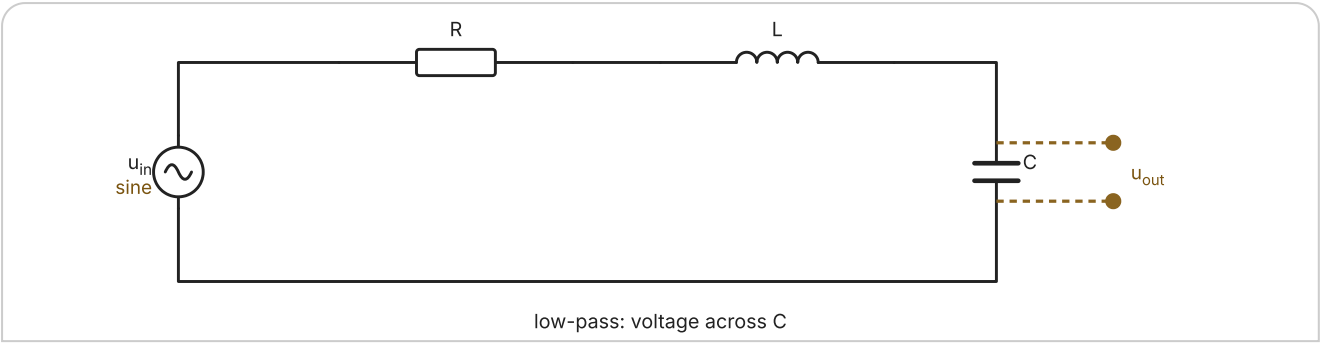
With sinusoidal signals differential equations become algebra. Charles Proteus Steinmetz introduced complex numbers into AC theory in 1893: a voltage $u(t) = \hat{U}\cos(\omega t + \varphi)$ is represented by the phasor $\underline{U} = \hat{U}e^{j\varphi}$, and differentiation becomes multiplication by $j\omega$. Each component then gets a complex impedance,

$$Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \frac{1}{j\omega C}$$

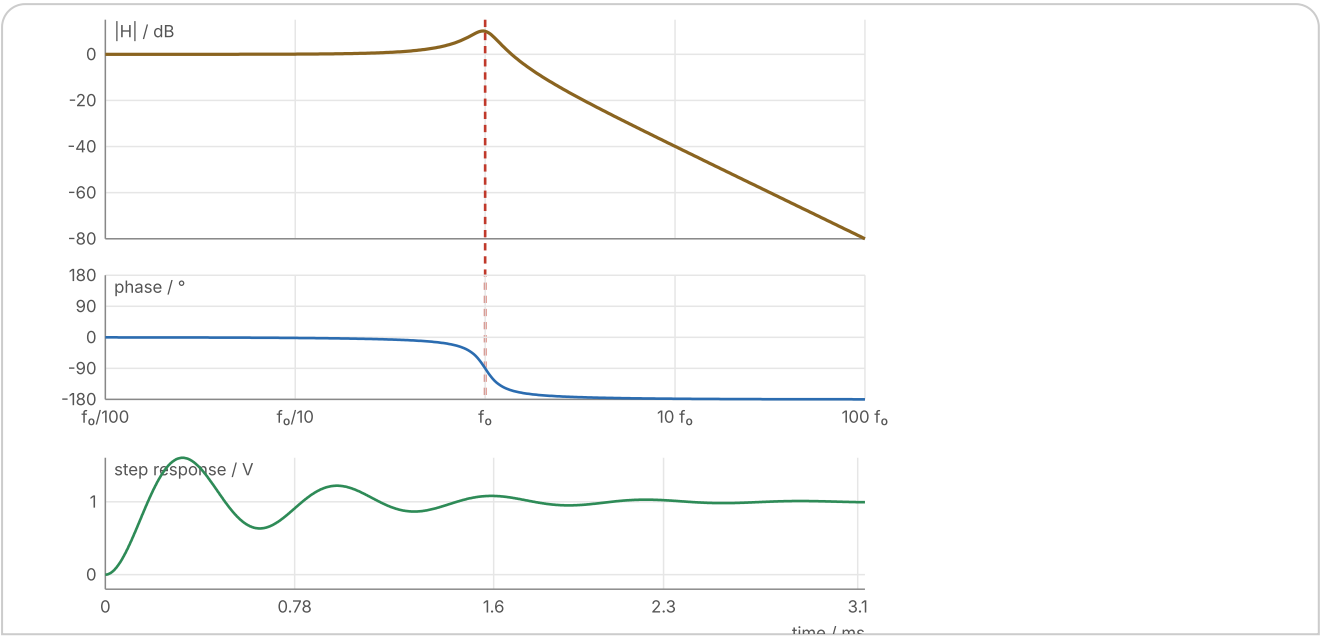
and Ohm's and Kirchhoff's laws hold unchanged for phasors and impedances. Engineers write j for $\sqrt{-1}$ because i was already taken by current. Steinmetz's method made alternating current calculable and was one of the reasons Westinghouse and General Electric won the "war of the currents" against Edison's direct current.

Resonance and filters

In a series RLC circuit the impedance is $Z = R + j(\omega L - 1/\omega C)$. At the resonance frequency $\omega_0 = 1/\sqrt{LC}$ the inductor and capacitor cancel, the impedance is smallest and the current largest. The quality factor $Q = \omega_0 L/R = 1/(2\zeta)$ tells how sharp the resonance is: the bandwidth is $\Delta f = f_0/Q$. Taking the voltage across the capacitor gives a low-pass filter; across the inductor a high-pass filter; across the resistor a band-pass filter. The frequency response is drawn in a *Bode plot* (Hendrik Bode, Bell Labs, 1930s) with logarithmic axes, on which the curves become almost straight lines: −40 dB per decade for a second-order low-pass filter above resonance.



low-pass: voltage across C

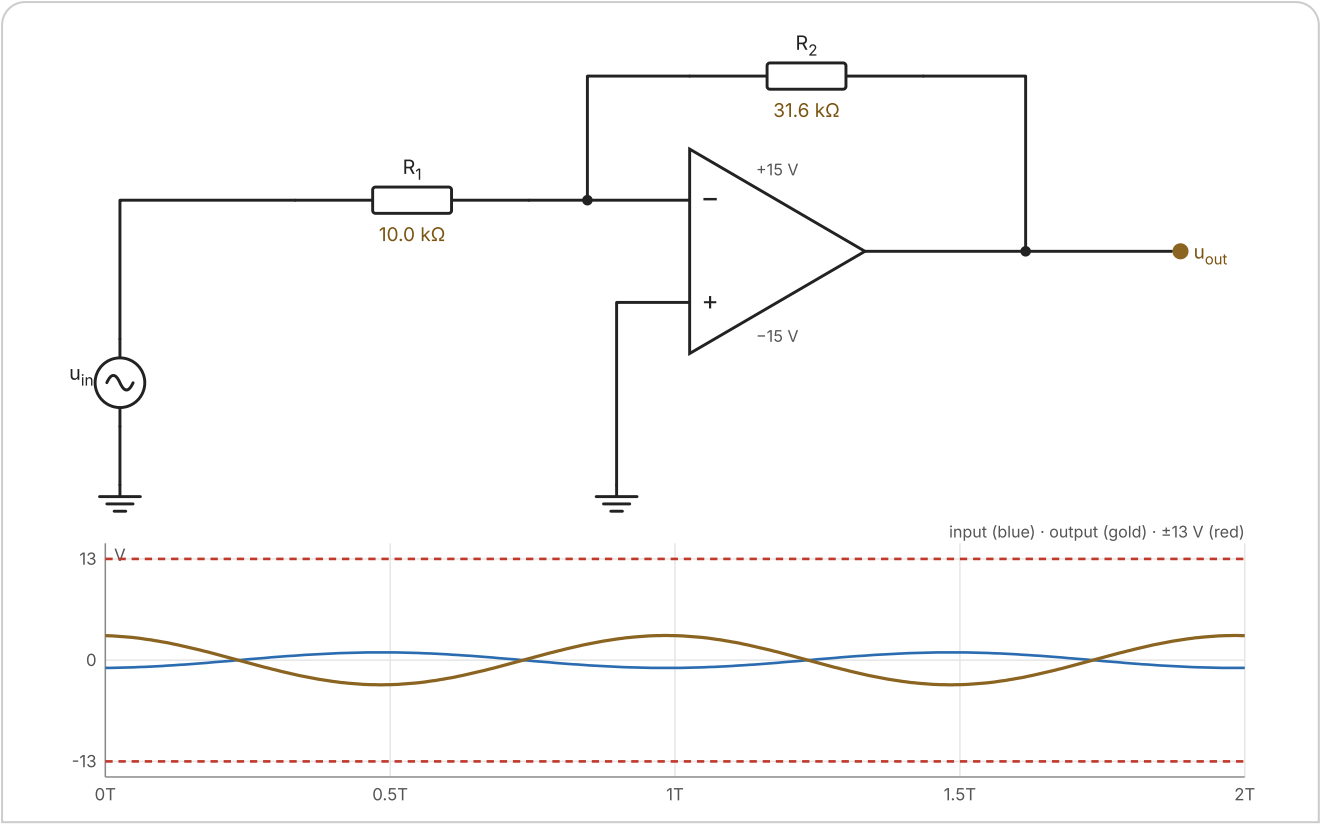


Power

With alternating current the power $p(t) = u(t)i(t)$ is not constant. Its mean, the active power, is $P = U_{\mathrm{rms}} I_{\mathrm{rms}} \cos\varphi$, where φ is the phase shift. An inductor or capacitor absorbs no power on average – the energy swings back and forth – but the current loads the wires all the same. That is why utilities require large motors to be compensated with capacitors so that the power factor $\cos\varphi$ is close to 1. The RMS value $U_{\mathrm{rms}} = \hat{U}/\sqrt{2}$ is the DC voltage that delivers the same power: the 230 V of the mains socket is the RMS value; the peak is 325 V.

The operational amplifier

The operational amplifier – a differential amplifier with enormous gain, originally built with valves for analogue computers – became the universal component of electronics with integrated circuits such as the µA741 (1968). With negative feedback the gain is set by two resistors alone: for an inverting amplifier $A = -R_2/R_1$. The two golden rules: no current flows into the inputs, and the feedback holds the voltage difference between them at zero. With capacitors in the feedback one builds integrators, differentiators and active filters without inductors.



The laboratory course

A laboratory course in circuit theory and electronics turns the theory into measurements, typically: RC time constants read on the oscilloscope, the resonance curve of an LC circuit measured point by point with a function generator and voltmeter, characteristics of diodes and transistors, a transistor amplifier with operating point and gain, and op-amp circuits. The most important lesson was often the same: the instrument affects what it measures – an oscilloscope input of 1 MΩ in parallel with 20 pF is not infinite.

Since 1989

- **Simulation:** SPICE is now free in programs such as LTspice, and circuits are designed and tested on screen before they are built.
- **Switch-mode:** Heavy 50 Hz transformers in power supplies have been replaced by switch-mode supplies working at 100 kHz and above – with gallium-nitride (GaN) transistors even in

small chargers.

- **Surface mounting:** Components have shrunk from through-hole resistors to SMD parts of 0.4 × 0.2 mm.
- **Cheap instruments:** A multi-channel digital oscilloscope with FFT now costs less than a textbook did in 1985.

Further reading

- Paul Horowitz and Winfield Hill: *The Art of Electronics* (3rd ed., Cambridge 2015)
- Charles K. Alexander and Matthew N. O. Sadiku: *Fundamentals of Electric Circuits* (7th ed., McGraw-Hill 2021)

- On this site: Electromagnetism · Applied field theory

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